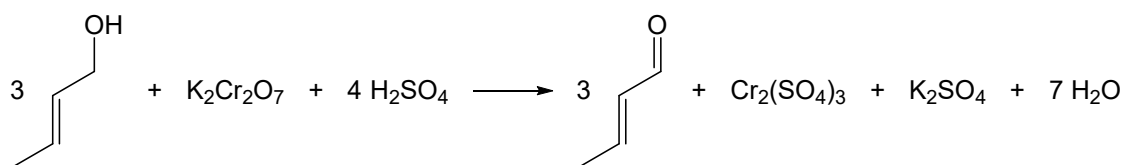


Section A**1(a)(i)**

Accept molecular formula in place of structural formula.

1(a)(ii)

Compound	M_r
$\text{C}_4\text{H}_6\text{O}$	70
H_2O	18
$\text{Cr}_2(\text{SO}_4)_3$	392.3
K_2SO_4	174.3

Atom economy for oxidation

$$= (3 \times 70) / (3 \times 70 + 392.3 + 174.3 + 7 \times 18) \times 100\% = 23.3\%$$

(allow ecf from **(a)(i)**)**1(a)(iii)**

Atom economy for a process can be inflated by having a larger molecule as the reactant/product, as seen in the horizontal trends of Table 1.1. So it is not valid. OWTTE.

Atom economy is valid for comparing between chemical processes when the same substrate is used, as seen from the vertical trends of Table 1.1. OWTTE.

1(a)(iv)

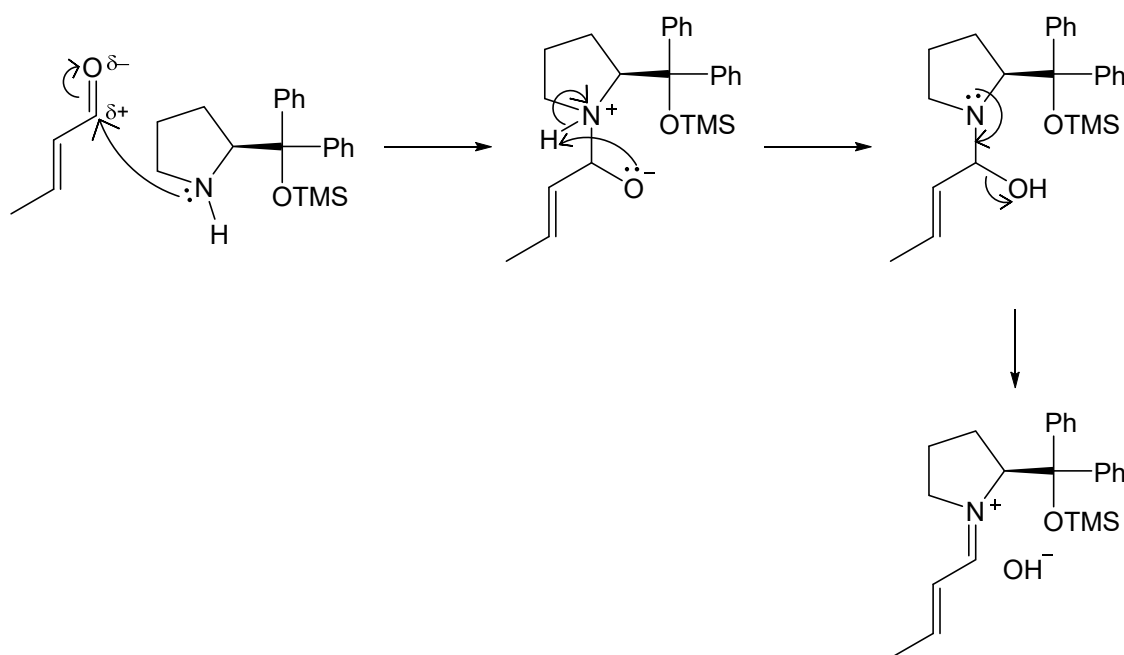
Atom economy does not take into consideration energy required to heat or cool reaction.

OR

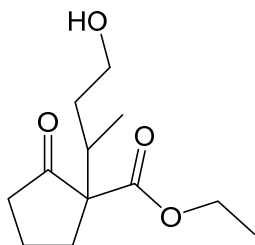
Atom economy does not take into consideration solvents used (which may not be recycled).

OR

Other reasonable answers that show negative impact on environment are accepted.

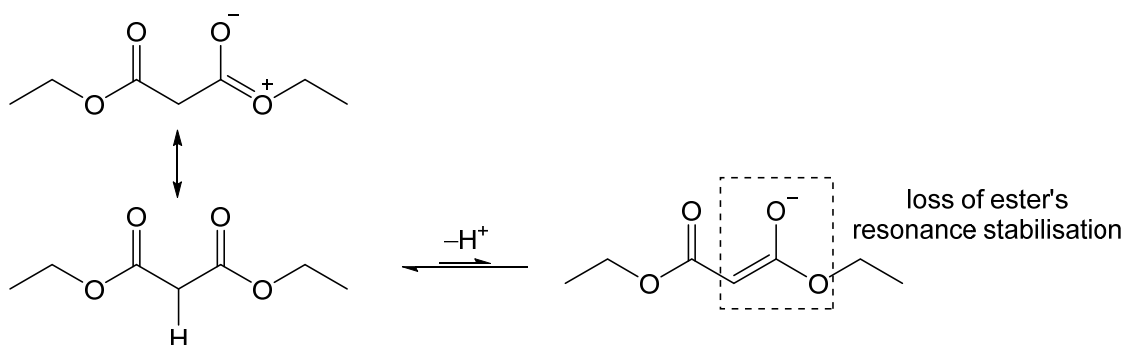
1(b)(i)

1(b)(ii)



1(b)(iii)

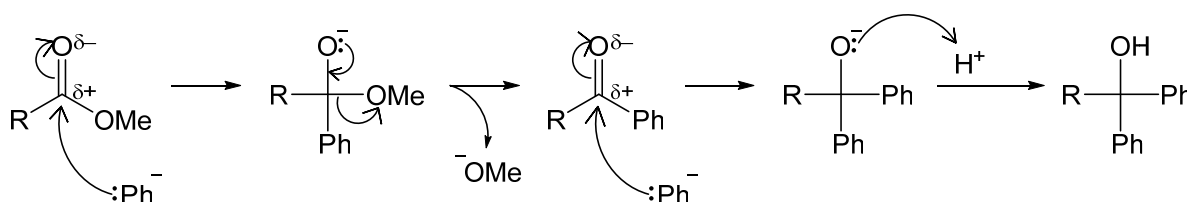
The conjugate base of diethyl malonate is less stable despite the electron withdrawing ester groups because the ester functional group itself already has resonance/delocalisation between O-C-O. There is less enolate stabilisation, as that would disrupt the resonance stabilisation of the ester functional group. In **X**, there is a ketone functional group so the formation of an enolate would not lead to a loss of any stabilisation already present in **X**.



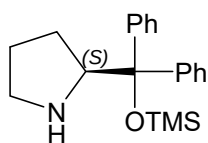
1(c)(i)

The Ph⁻ nucleophile can also act as a base and deprotonate the H on the amine, leading to lowered yield, as PhMgBr would be consumed.

1(c)(ii)



1(d)(i)



1(d)(ii)

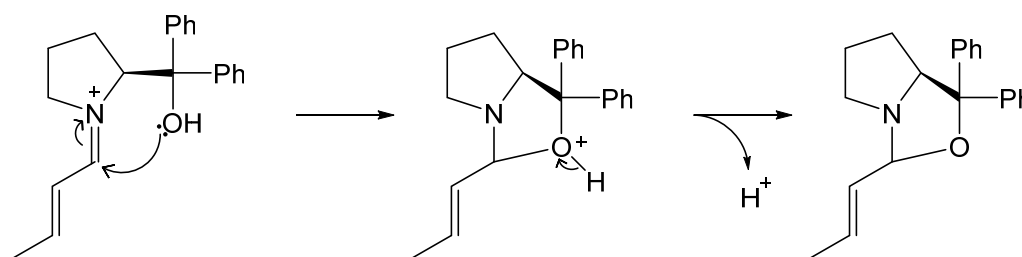
$$84\% = (-9.3 / [\alpha]_{\text{pure}}) \times 100\%$$

$$[\alpha]_{\text{pure}} = -11.1^\circ$$

1(d)(iii)

Carbonyl (C=O) and/or ester (COO) functional groups on nucleophile can interact with OH on non-silylated organocatalyst via hydrogen bond, causing some nucleophiles to approach from the "top", rather than from the "bottom" due to steric hindrance. This causes a drop in enantioselectivity.

1(d)(iv)

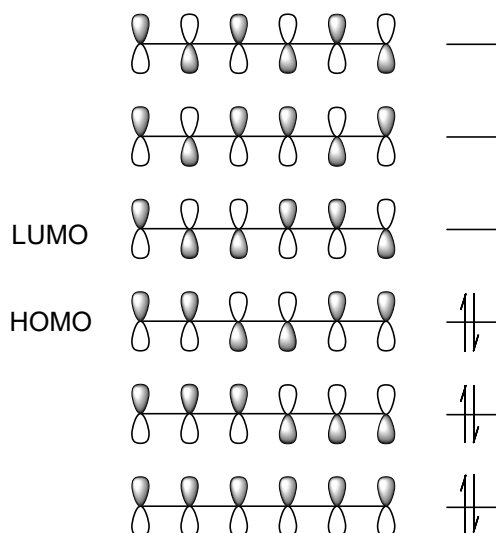


- 1(d)(v)** Organocatalyst is greener because:
- Natural, renewable sources (e.g. amino acids)
 - Some metals are toxic

Accept any other reasonable answers related to Green Chemistry

2(a)(i) $\pi \rightarrow \pi^*$

2(a)(ii)



2(a)(iii) π bond order before = $(6 - 0) / 2 = 3$
 π bond order after = $(5 - 1) / 2 = 2$

2(a)(iv) When *trans*-hexa-1,3,5-triene absorbs a photon that matches the $\pi \rightarrow \pi^*$ energy gap, the π bond order is lowered and the conjugated π bond is weakened / the π bond is broken. This reduces the restriction of rotation about the double bond, allowing for rotation about the σ bond.

2(b)(i) Z

2(b)(ii) β -Carotene has a highly conjugated π system, so the $\pi \rightarrow \pi^*$ energy gap narrows and corresponds to energy of wavelength of visible light.

2(b)(iii) When carrots are soaked in aqueous bromine, the alkenes in β -Carotene will react with aqueous bromine, which leads to a decrease in the extent of conjugation.

Hence, this increases the $\pi \rightarrow \pi^*$ energy gap and the molecule will absorb at (shorter) wavelengths outside of the visible light range.

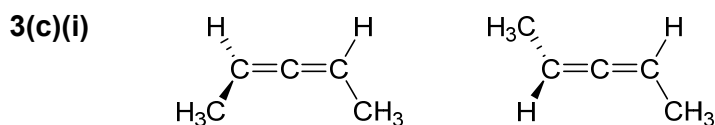
3(a)(i) Absorption of IR radiation causes stretching or bending vibrations in covalent bonds in a molecule [1], which leads to a change in dipole moment.

3(a)(ii) C1: sp^2 ; C2: sp ; C3: sp^2

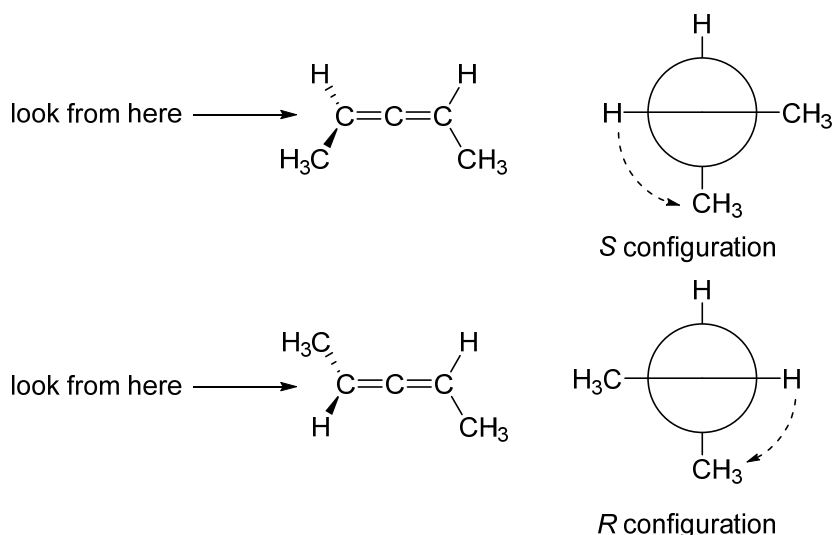
3(a)(iii) Alkene C=C wavenumber is $1635 - 1690\text{ cm}^{-1}$. Since allene's absorption is at a higher wavenumber of 1960 cm^{-1} (i.e. higher energy), it implies that the bond strength is stronger.

This is because the σ bond forming the C=C bond in an allene is formed from an $sp-sp^2$ overlap while that for an alkene is formed from an sp^2-sp^2 overlap. Since the sp orbital has greater s-character, it is less diffuse and the $sp-sp^2$ overlap is more effective.

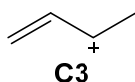
3(b) The circulation of π electrons near the acetylenic proton induces a magnetic field that opposes the applied field that the molecule is placed in. The proton is shielded and requires a lower frequency to reach resonance. Hence, a lower δ value.



3(c)(ii)

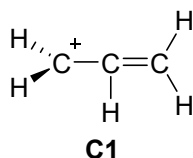


3(d)(i)



C3 is an allylic carbocation that is stable as the empty p-orbital on C^+ overlaps with the adjacent π electron cloud and the electrons from the π bond can delocalise and disperse the positive charge.

3(d)(ii)



C1 is less stable than **C3**. As the two π electron clouds are 90° to each other in the allene, so the empty p-orbital of the resulting carbocation is also 90° to the adjacent π electron cloud before rotation, so there is no overlap of p-orbitals, no delocalisation and no dispersal of positive charge, unlike an allylic carbocation. (Note: this carbocation is about as stable as a primary carbocation)

3(d)(iii) Hammond's Postulate states that the transition state most resembles the adjacent reactant, intermediate or product that it is closest in energy to, as long as the energy difference between the transition state and the adjacent structure is not too large.

Since the reaction is endothermic, transition state most resembles the carbocation.

TS 1 would resemble the less stable C1, so TS 1 is also less stable (i.e., higher in energy).

4(a)(i) Br at the 2-position would experience some destabilisation due to steric hindrance with the methyl group at 1-position. Hence, H_2 is at a higher energy level than H_4 and thus closer to H_1 . So $\Delta H_d = H_x - H_1$ for Br at 2-position is less exothermic. Br at the 4-position does not experience such steric hindrance.

4(a)(ii) $\Delta H_r = \Delta H_1 + \Delta H_d$

4(a)(iii)
$$\begin{aligned} E_a &= A + B\Delta H_r \\ &= A + B(\Delta H_1 + \Delta H_d) \\ &= A + B\Delta H_1 + B\Delta H_d \end{aligned}$$

Since ΔH_1 is a constant, $A + B\Delta H_1$ is also a constant.

$E_a = A' + B\Delta H_d$ (Linear relationship, shown)

4(b) Let $E_a = A' + B\Delta H_d$
From first two lines of the table,
 $95.9 - 84.3 = B[1.75 - (-14.7)]$
 $B = 0.705$

Then $A' = 84.3 - 0.705(-14.7) = 94.7$

So E_a for 4-position = $94.7 + 0.705(-19.4) = +81.0$

4(c) The mechanisms for the nitration at various positions on the benzene ring are not similar so the reactions are not closely-related. OWTTE

5(a)

Number of C atoms in **E** = $\left(\frac{100}{1.1}\right) \left(\frac{6}{55}\right) = 9.9 \approx 10$

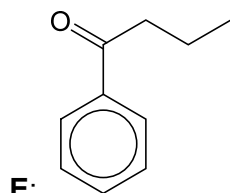
Since the M_r of **E** = 148 and it contains 12 hydrogen atoms,

Number of oxygen atoms = $[148 - 12(1.0) - 10(12.0)] \div 16 = 1$

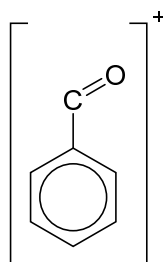
Therefore, the molecular formula of **E** is $C_{10}H_{12}O$.

5(b)(i)

δ/ppm	Integration	Splitting	Deductions
1	3	Triplet	CH_3 group adjacent to a CH_2 group / Presence of $-\text{CH}_2\text{CH}_3$
1.8	2	Multiplet	CH_2 group next to two sets of non-equivalent protons / Possibly presence of $-\text{CH}_2\text{CH}_2\text{CH}_3$
2.9	2	Triplet	Protons are deshielded $\rightarrow$ near an electron withdrawing $\text{C}=\text{O}$ group CH_2 group adjacent to a CH_2 group / Presence of $\text{CO}-\text{CH}_2\text{CH}_2-$
7.3	5	Multiplet	5 aromatic protons of a mono-substituted benzene



5(c)(i)



5(c)(ii)

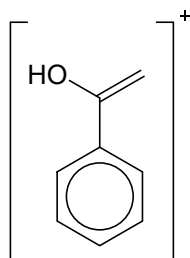
The fragment is very stable due to resonance as the positive charge is on the benzylic carbon atom and also on the $\text{C}=\text{O}$ carbon atom. It has a higher chance of reaching the detector in the mass spectrometer before it breaks down further, so the relative intensity is the high.

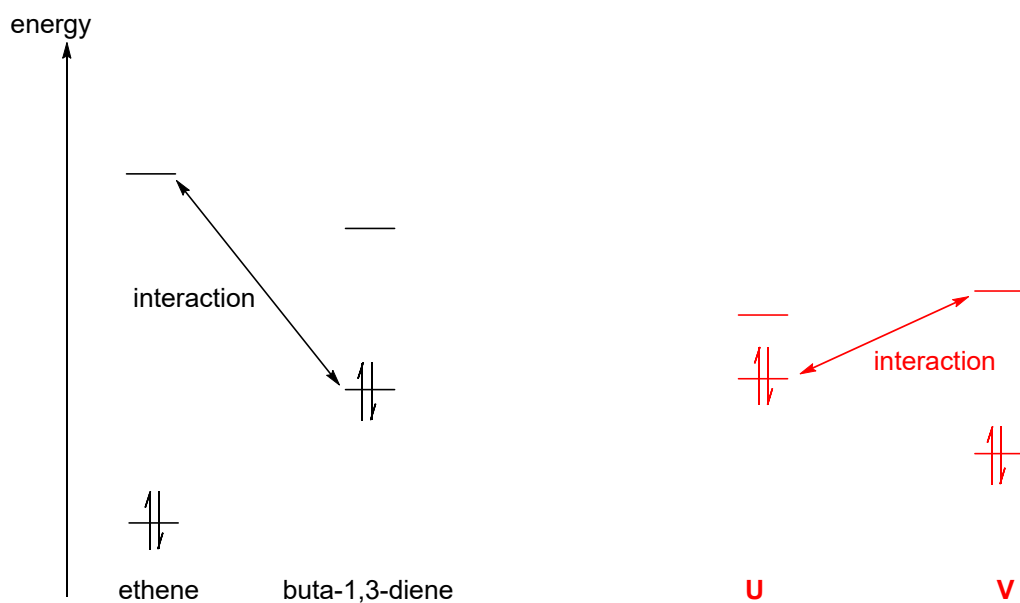
OR

Longer lifespan in the spectrometer OWTTE

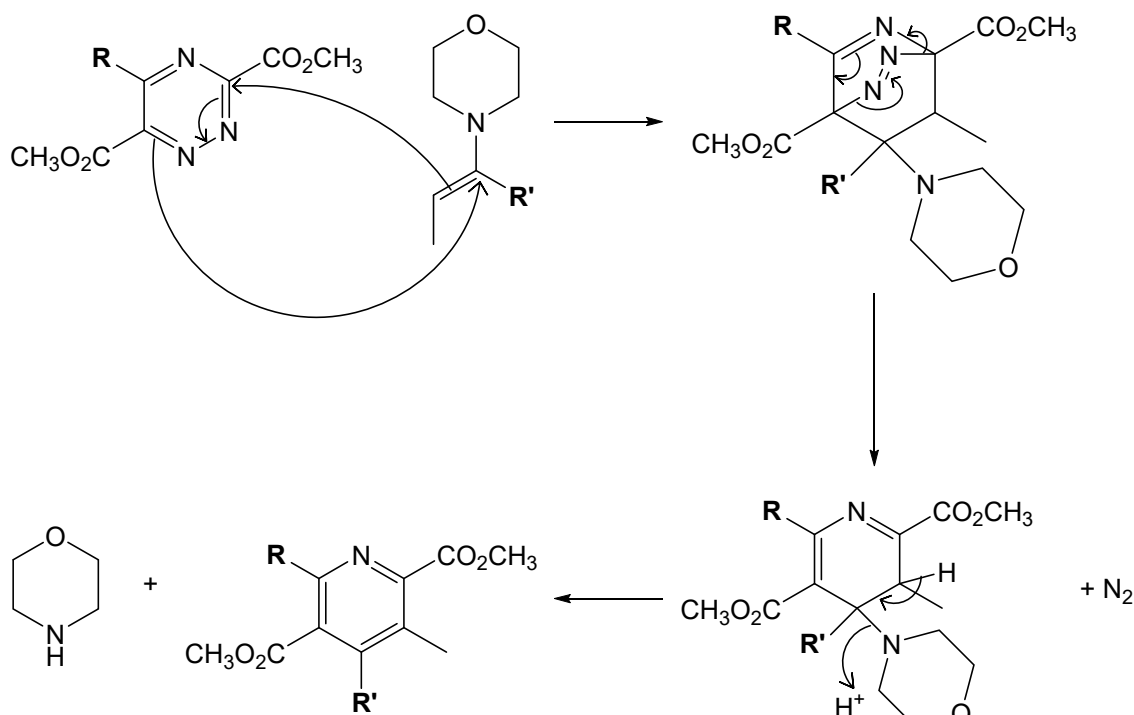
5(d)

McLafferty Rearrangement

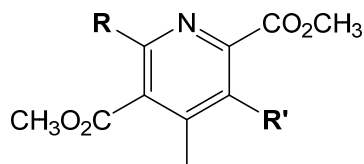


Section B**6(a)(i)****6(a)(ii)**

A smaller energy gap between the HOMO of **U** and LUMO of **V** results in a better overlap/interactions of orbitals[1] in the transition state, stabilising the transition state and increasing the rate of reaction between **U** and **V**.

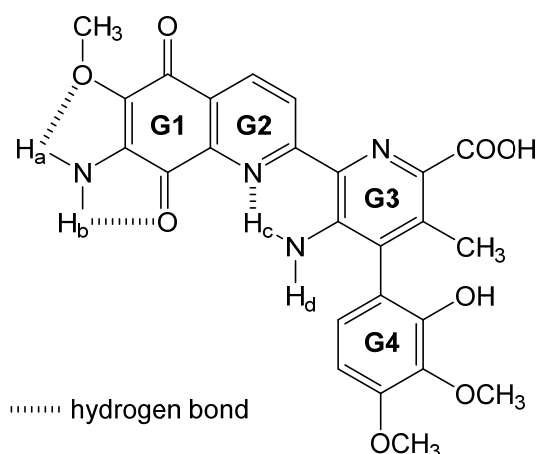
6(b)(i)

6(b)(ii)



6(c)(i)

Both H_a and H_b are able to form intramolecular hydrogen bonding with the oxygen atom of $-OCH_3$ and $C=O$ on ring **G1** respectively via a 5-membered ring. H_d is unable to form any intramolecular hydrogen bonding whereas H_c is able to form strong intramolecular hydrogen bonding with the N on ring **G2** via a 6-membered ring. This intramolecular hydrogen bonding involving H_c is much stronger than that involving H_a/H_b as N atoms is in much closer proximity via a 6-membered ring compared to the 5-membered ring involving H_a/H_b and the oxygen atom. Due to the much stronger intramolecular hydrogen bonding, H_c is much more deshielded than H_d , resulting in a large difference in the proton chemical shift values.



6(c)(ii)

signal label	chemical shift δ / ppm	splitting pattern	integration value	identity of protons
S1	2.18	singlet	3H	$-CH_3$
S2	3.16 – 3.20	–	9H	$-OCH_3$ on rings G1 and G4
S4	6.70 – 6.73	pair of doublets	2H	Ar-H on ring G4 (splitting due to coupling to nearby proton)
S7	8.36 – 8.92	pair of doublets	2H	Ar-H on ring G2 (splitting due to coupling to nearby proton; more deshielded due to N present on ring)
S8	9.01	singlet	1H	$-OH$
S10	12.32	Singlet	1H	$-COOH$

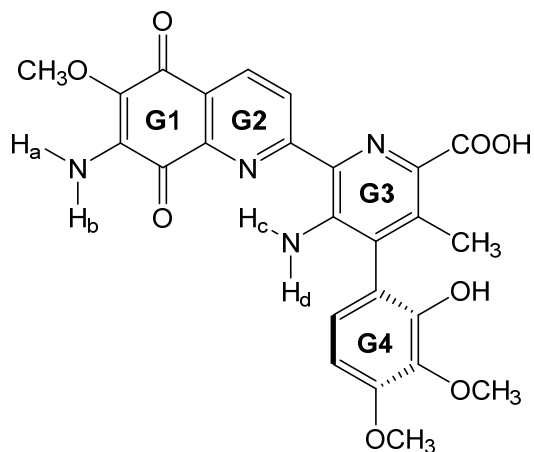
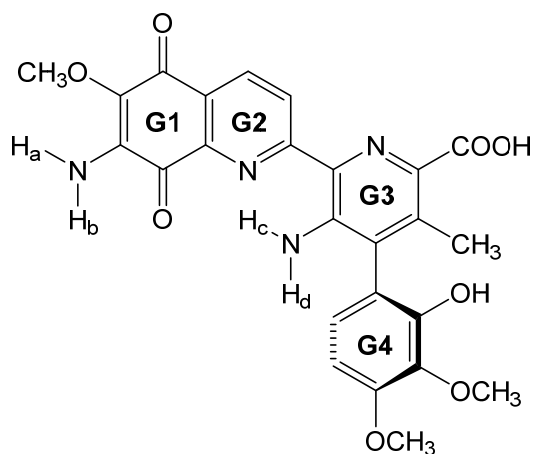
6(d)(i)

The signals of streptonigrin and this internal standard do not overlap with each other.

6(d)(ii)

$$\%P_{\text{sample}} = 0.280 \times \frac{9}{3} \times \frac{2.00}{5.00} \times \frac{506.5}{224.3} \times 99.95 = 75.8\%$$

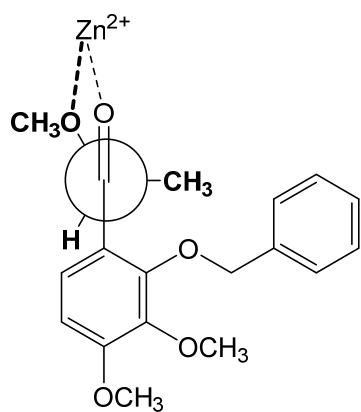
6(e)



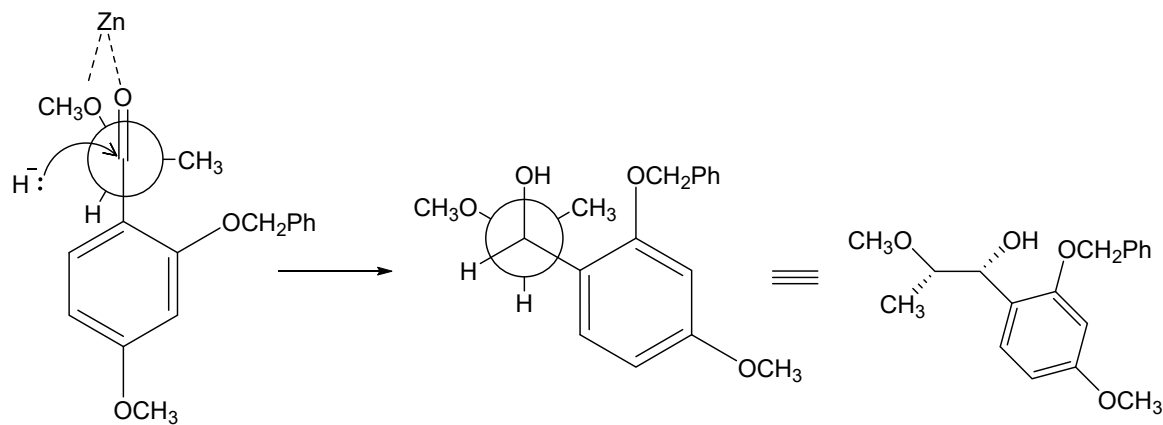
Rings **G3** and **G4** are not in the same plane / are perpendicular to each other as there would be steric hindrance caused by the $-\text{CH}_3$ and $-\text{OH}$ groups respectively.

Hence, streptonigrin exhibits enantiomerism.

6(f)(i)

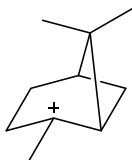


6(f)(ii)(iii)



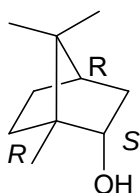
The coordination to Zn^{2+} “locks” the conformation in place. The large phenyl group on the right hinders the approach of the H^- nucleophile, so H^- attacks from the left.

7(a)(i)

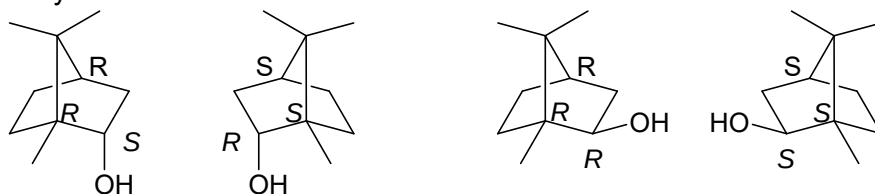


Although this intermediate is a tertiary carbocation, it has a 4-membered ring in the structure, resulting in significant ring strain.

7(a)(ii)

7(a)(iii) 4 stereoisomers of borneol

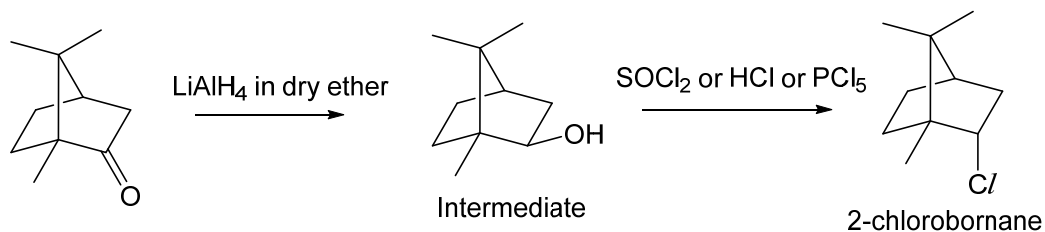
They are:



Note: While there are three chiral centres, the two chiral centres at the bridgehead positions have their configurations “synced” to the each other. They will always be R, R or S, S.

Hence, the total number of stereoisomers = $2^{1+1} = 2^2 = 4$

7(a)(iv)

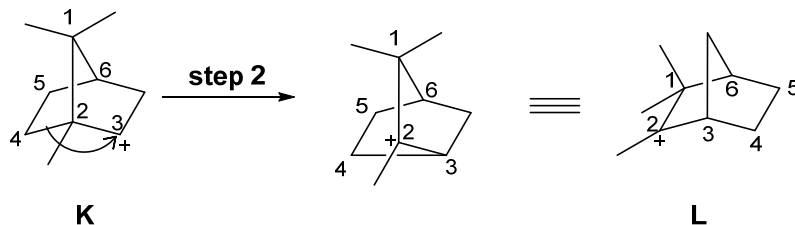


7(a)(v)

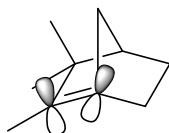
The molecular ion peak for camphor has a m/e value of 152, while m/e value for 2-chlorobornane is 172.

2-chlorobornane would have M and M+2 peaks, with relative intensity of 3:1.

7(b)(i)

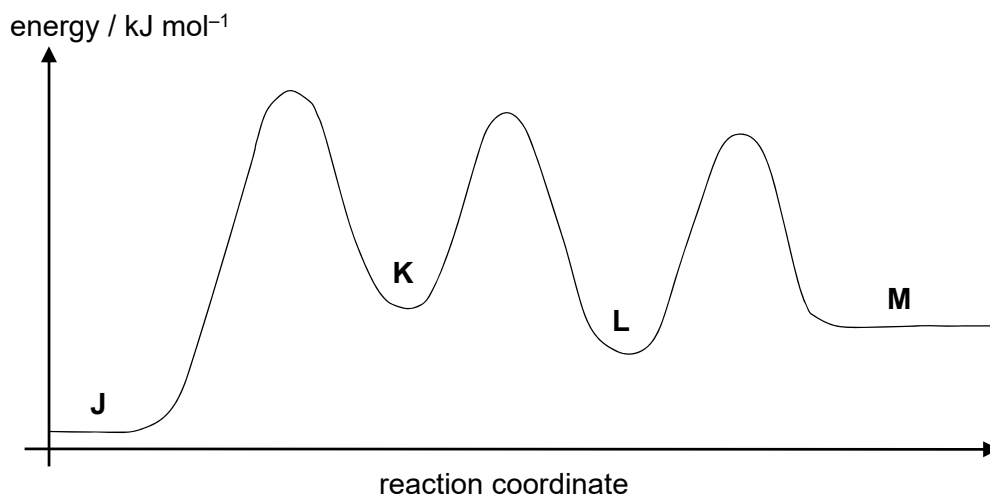


7(b)(ii)

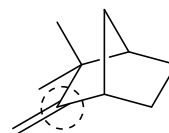


The Zaitsev's product cannot exist because the p orbitals of the bridgehead atom and adjacent atoms are orthogonal and thus are not aligned properly for the overlap to form pi bonds.

7(b)(iii)



Note: **M** is higher in energy than **J** because **M** is more unstable due to the double bond. The trigonal planar (120° angle) is difficult to achieve at the carbon atom circled in the molecule and there is some strain.



7(b)(iv)

$$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$$

Elimination reactions has a more positive entropy change compared to substitution reaction due to greater number of products formed.

The higher the temperature, the more positive the $T\Delta S^\ominus$, the more negative the $\Delta G^\ominus$. Hence the yield of the eliminated products increases.

7(b)(v)

Use a stronger/sterically hindered base such as DBU/DBN.

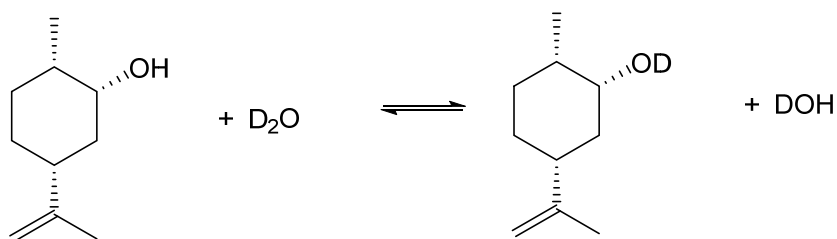
Accept other chemically sensible answer.

7(c)(i)

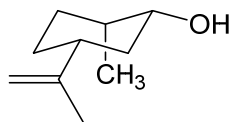
Any 2 differences

1. There are 3 signals due to the methyl groups in camphor while there are 2 signals due to the methyl groups in isodihydrocarveol
2. All the signals due to the methyl groups in camphor are singlets while in isodihydrocarveol, one of the signal for the methyl group is a doublet.
3. There is a signal due to the hydroxyl proton in isodihydrocarveol but not in camphor.
4. There are 2 signals due to the $=CH_2$ in isodihydrocarveol, and both signals exist as doublets. However, these signals are absent in camphor.

7(c)(ii) There is a rapid proton exchange between the hydroxyl proton and D₂O.



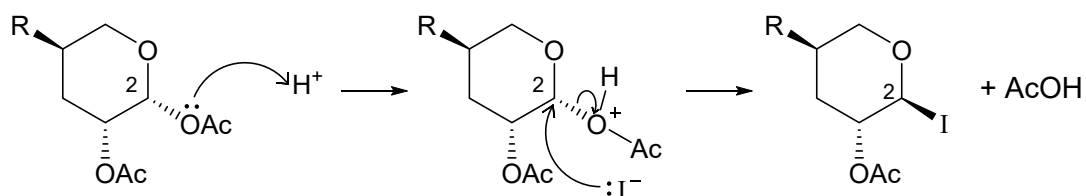
7(c)(iii)



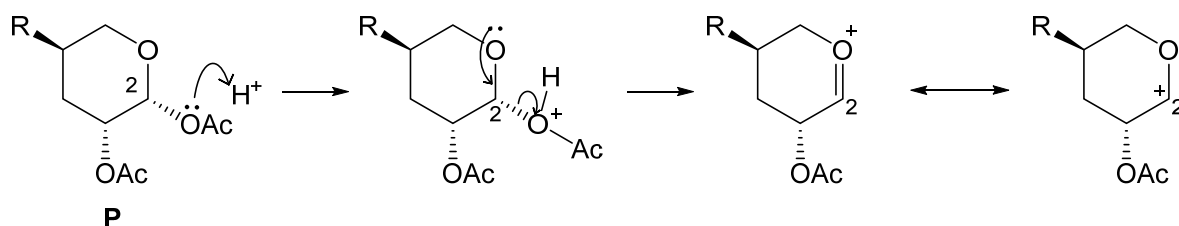
7(c)(iv)

The most stable chair conformation of isodihydrocarveol does not have the hydroxyl group in the axial position. Hence it will need to undergo a ring flip to the more unstable conformation so as to achieve anti-periplanar geometry for E2 elimination to occur.

8(a)(i)



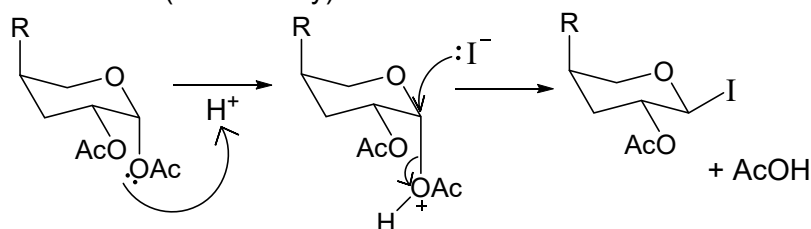
8(a)(ii)



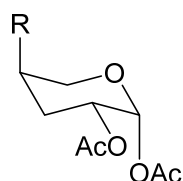
Presence of O adjacent to carbocation leads to resonance stability. Neighbouring group effect.

8(a)(iii)

Mechanism (for info only):

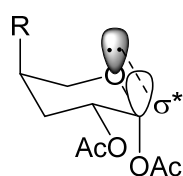


Backside attack for S_N2 requires this conformer:



Since 2 of the substituents (including the large R group) are axial, there would be significant 1,3-diaxial interactions which leads to instability.

8(a)(iv)



Hyperconjugation of the empty σ^* of C2 and the axial non-bonding lone pair electrons of the ring O atom stabilises the structure, despite the steric strain. Anomeric effect.

8(a)(v)

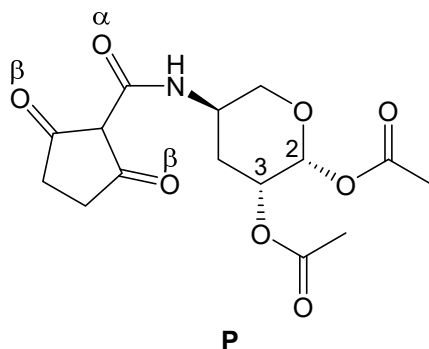
Acetoxy on C3 will be equatorial and cannot undergo S_N2 since the nucleophile that approaches from the backside would be hindered by the ring.

S_N1 is not desirable under the conditions as the carbocation formed is not so stabilised (lack of neighbouring group effect).

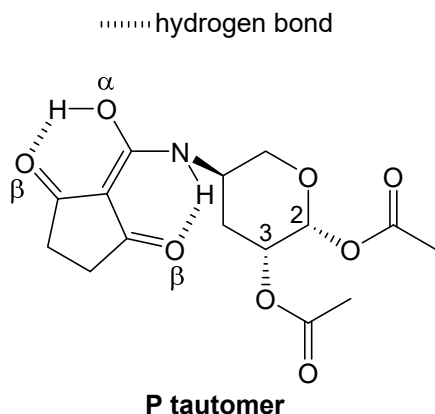
8(b)(i)

$\nu_{\max} / \text{cm}^{-1}$	strength	appearance	assignment
3450	w	very broad	N-H
3300	m	very broad	Enolic O-H of αO Hydrogen bonded
1750	s	sharp	AcO C=O (ester)
1691	s	-	βO C=O
1634	s	-	αO C=O

8(b)(ii)



The most stable tautomer of **P** is actually:



Justifications

The ring C=Os have a lower wavenumber IR signal than usual since the O atoms form hydrogen bond, reducing electron density and bond order from C=O. They also form tautomers (the 3 tautomers are in an equilibrium mixture with **P**)

The amide C=O has much lower wavenumber IR signal, as it is significantly weakened to bond order < 2 , and the tautomer tends to form.

This signal shifts when αO is replaced with ^{18}O . Since heavy isotope results in larger mass, smaller vibrational energy, and down shift in wavenumber

There is an O-H signal from the enol. This signal shifts when αO is replaced with ^{18}O . Since heavy isotope results in larger mass, smaller vibrational energy, and down shift in wavenumber.

8(c) $[R]$ at beginning of reaction = $A/\epsilon l = 1.5/[12750(1.0)] = 0.0001176 \text{ mol dm}^{-3}$.

Let $x = [S]$ at end of reaction.

Therefore $[R]$ at end of reaction = $0.0001176 - x$

$$1.09 = 12750(0.0001176 - x)(1.0) + 6500(x)(1.0) = 1.5 - 12750x + 6500x$$

$$6250x = 0.41$$

$$x = 0.0000656 \text{ mol dm}^{-3}$$

$$\% \text{ yield} = 0.0000656/0.0001176 \times 100\% = 55.8\%$$

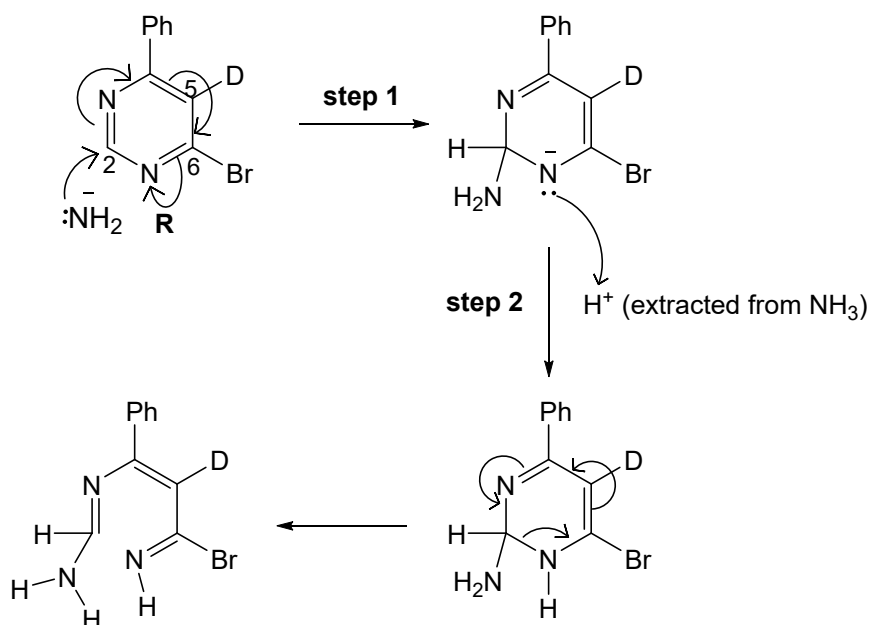
8(d)(i)

C2 is less hindered than C6.

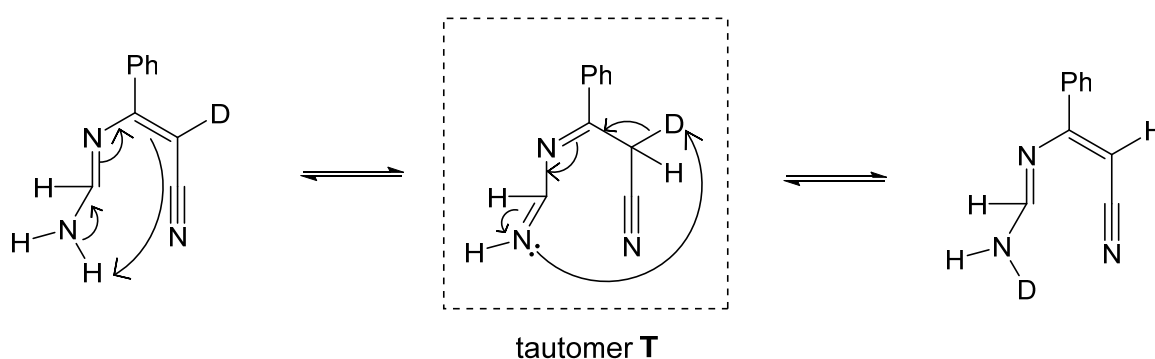
OR

C2 has lower electron density/(more δ^+)/more electron deficient than C6 as it is between two electronegative N atoms.

8(d)(ii)



8(d)(iii)



Note:

Fast tautomerism enables the D to exchange positions with the NH_2 protons, the only group with exchangeable protons.

The D can then be lost during exchange with solvent NH_3 protons.